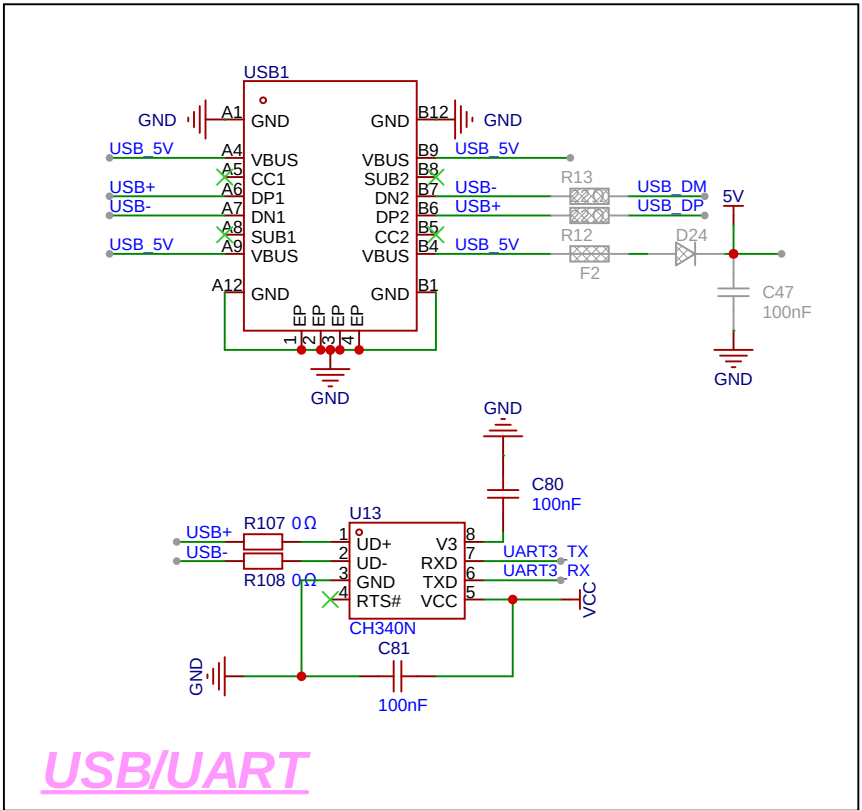
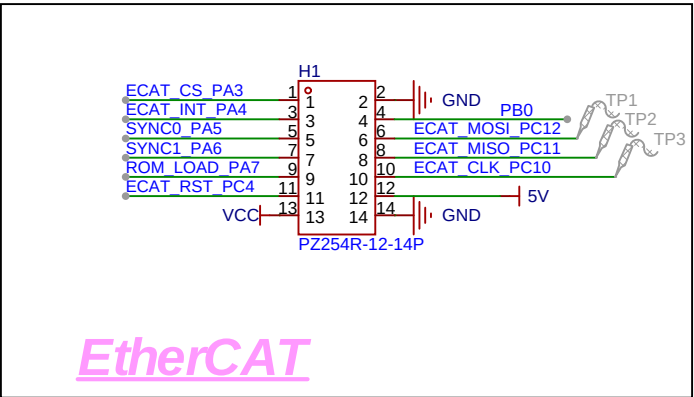
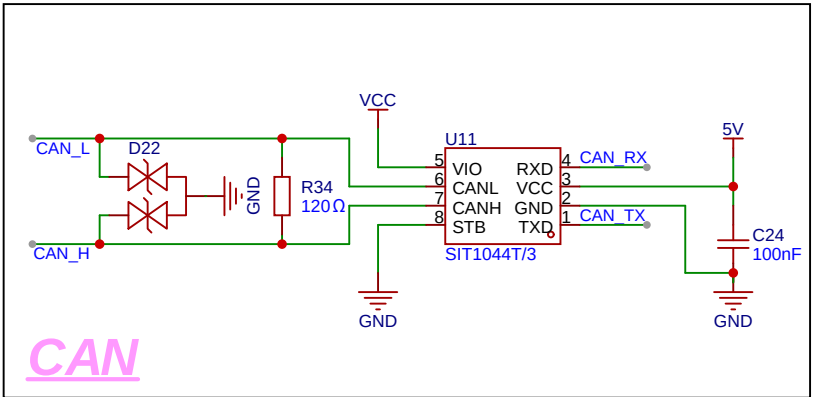
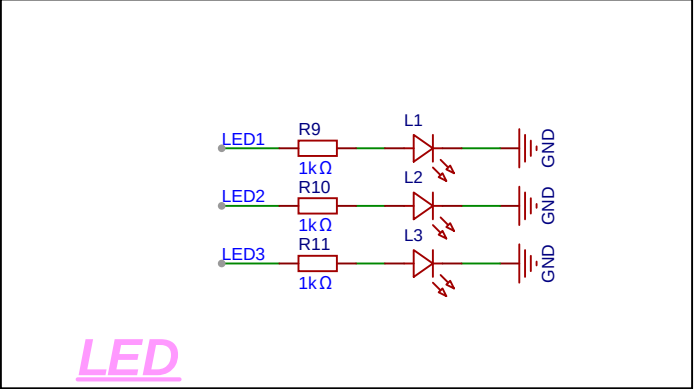
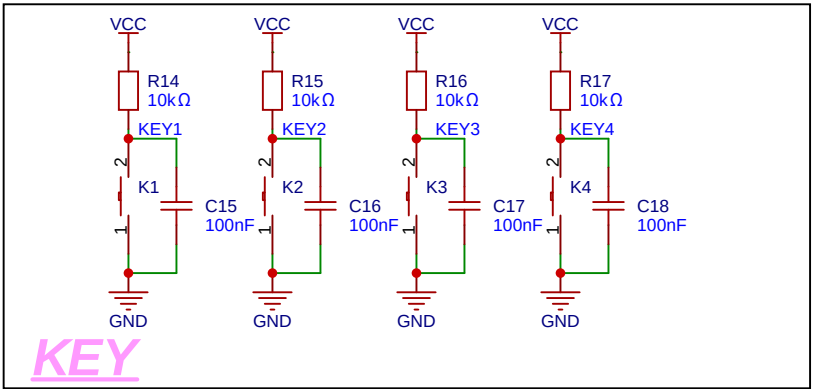
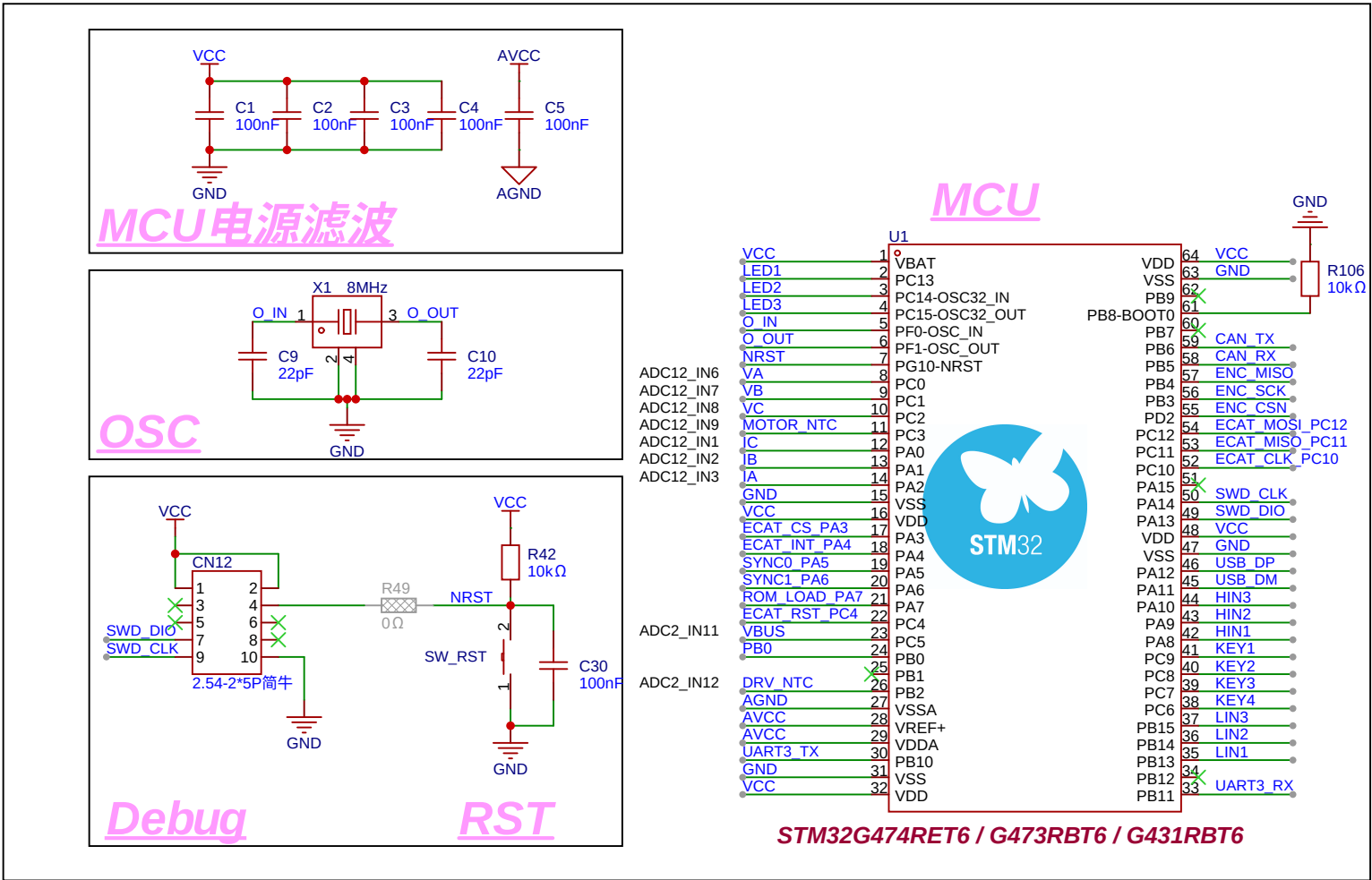
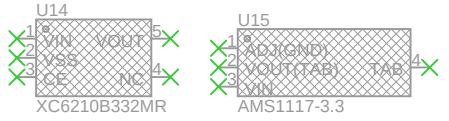
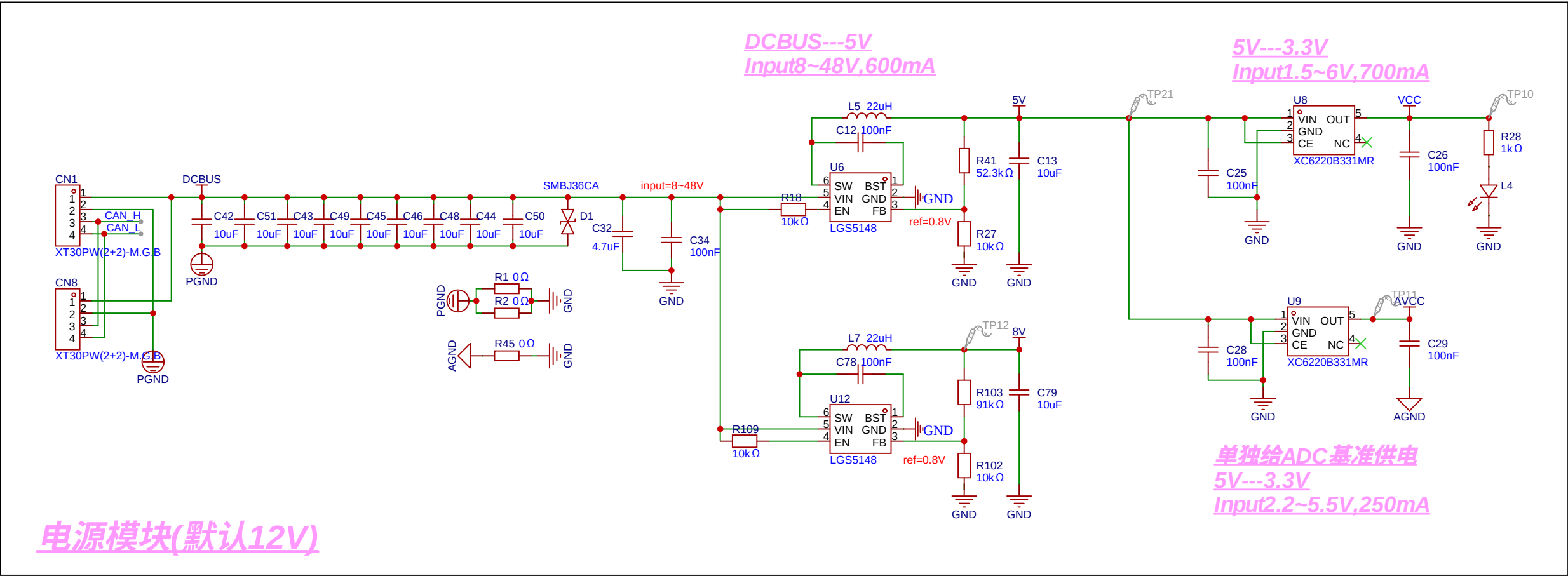
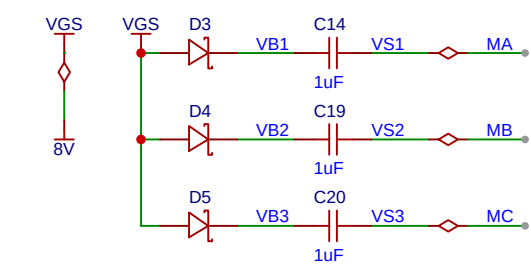


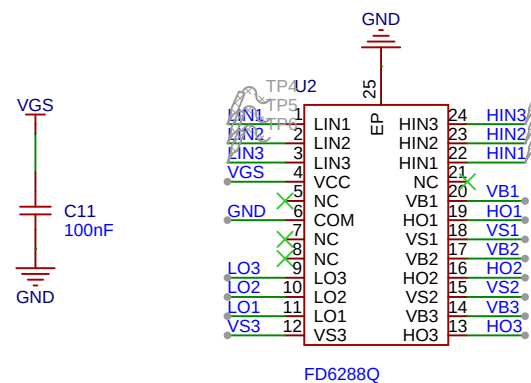




电源处理



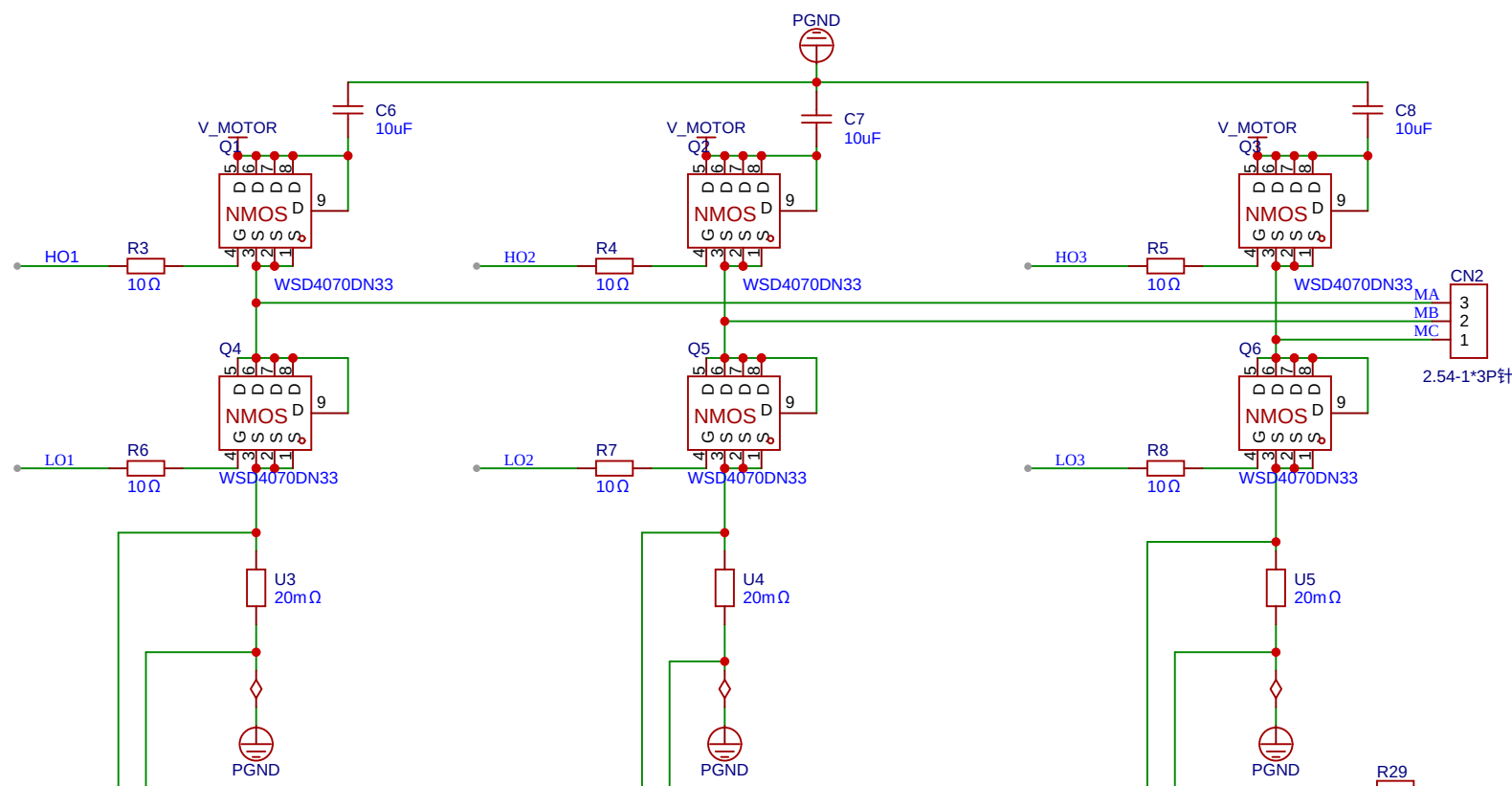
自举电路



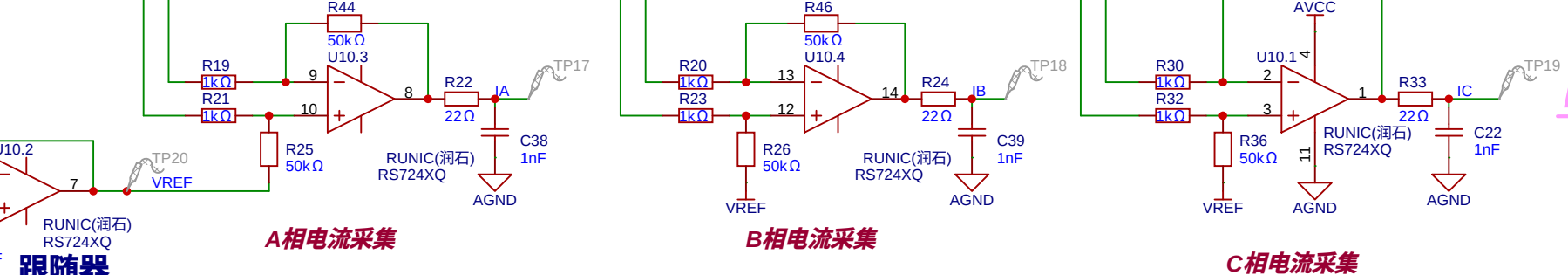
预驱芯片



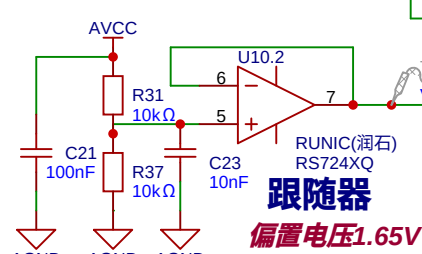
MOS驱动电路



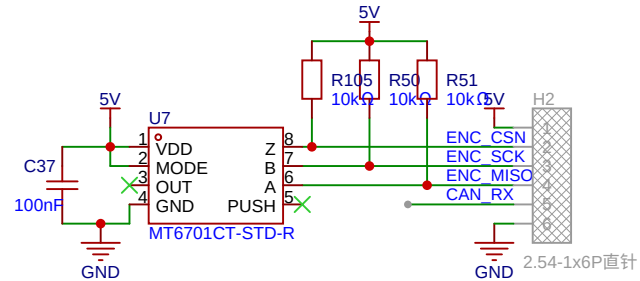
电流采样



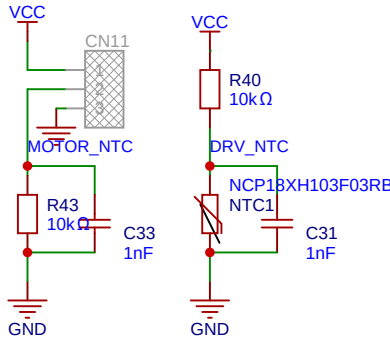
跟随器



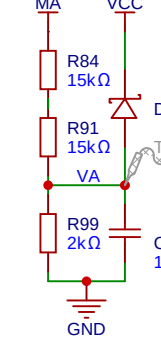
MT6701CT编码器



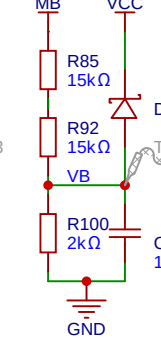
温度采样



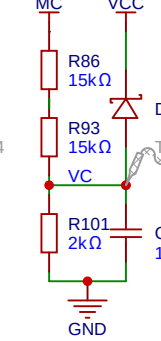
A相电压采样



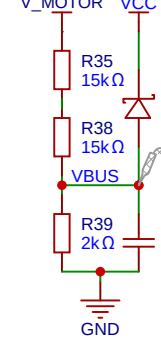
B相电压采样



C相电压采样



母线电压采样



TITLE:	iRDT_Driver	REV:	V1.3
Company:	机器人驱动小部落	Sheet:	3/3
Date:	2026-05-20	Drawn By:	Intelligent Robot-Driven Tribe

采样点最高电压为: 3.3V + 二极管管压降

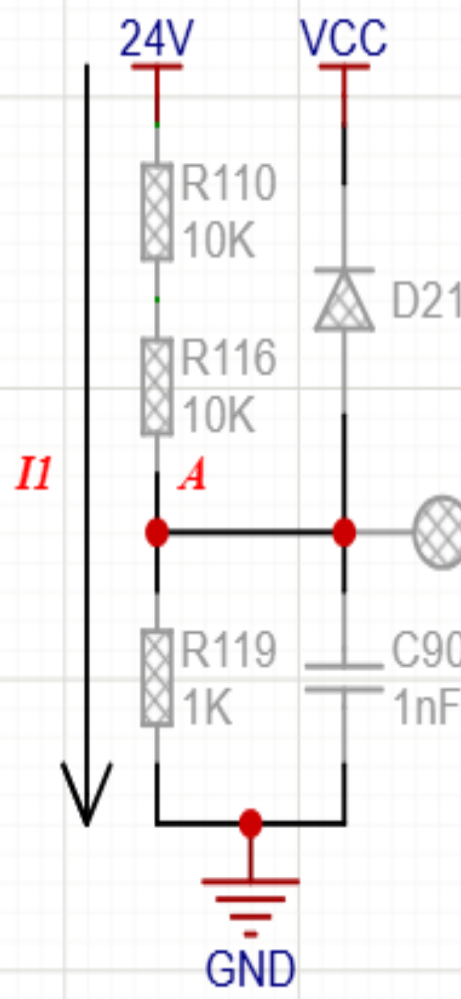
采样点电压应接近3.3V的一半,分辨率高, 易采样

I1电流: $24V / (10k + 10k + 1k) = 1.14mA$

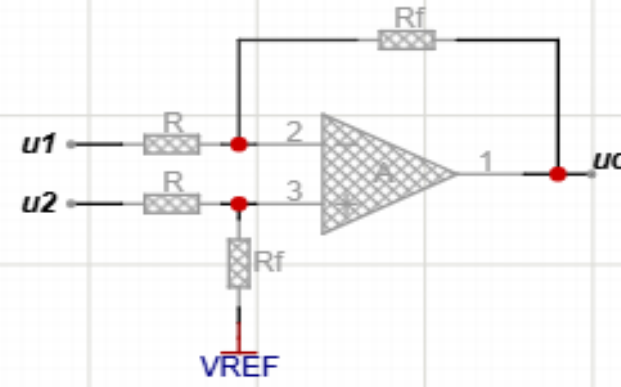
A点电压: $I1 * 1k = 1.14V$

缩小电压系数: $k = (10k + 10k + 1k) / 1k$

$VA = adc_val * (3.3V / 4096) * k$



电流采样计算



运放增益: $k = Rf / R$

偏置电压: $VREF = 1.65V = 2048$

采样电阻: $Rm = 1m\Omega$

运放输出: $uo = k * (u2 - u1)$

$u2 - u1 = I * Rm$

$uo = (adc_val - 2048) * 3.3V / 4096$

采集电流: $I = uo / (Rm * k)$